

HW3 · Shakespeare RNN Report

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0. Overview & Setup

這次的作業是RNN並搭配 Shakespeare corpus (train/valid 依提供檔)的資料。我的框架是：PyTorch + GPU AMP 混合精度，輸出統一在 plots/ , outputs/ , checkpoints/ 。亂數種子 113024510，**並且每個 batch 的 hidden state 依作業提示重設為 0**。

重點設定：

- Optimizer: Adam, lr=0.002, clip_grad_norm=5, batch=256
- Epochs: 5 (patience=5，保證 5 個 breakpoints)
- Model: RNN / LSTM，hidden size = [64, 128, 256] 與 seq_len = [50, 100, 150]，其中Baseline為 hs=128, seq=100。
- 評估：loss、BPC (=loss/ln2)、acc、err

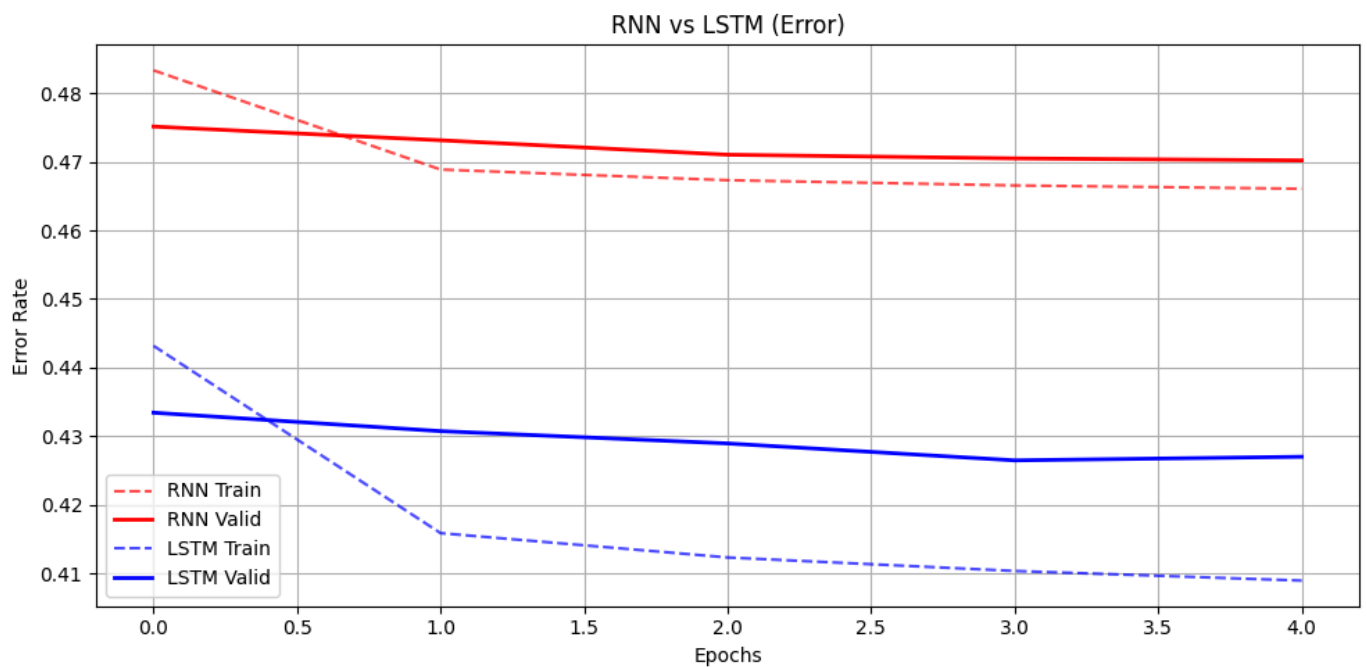
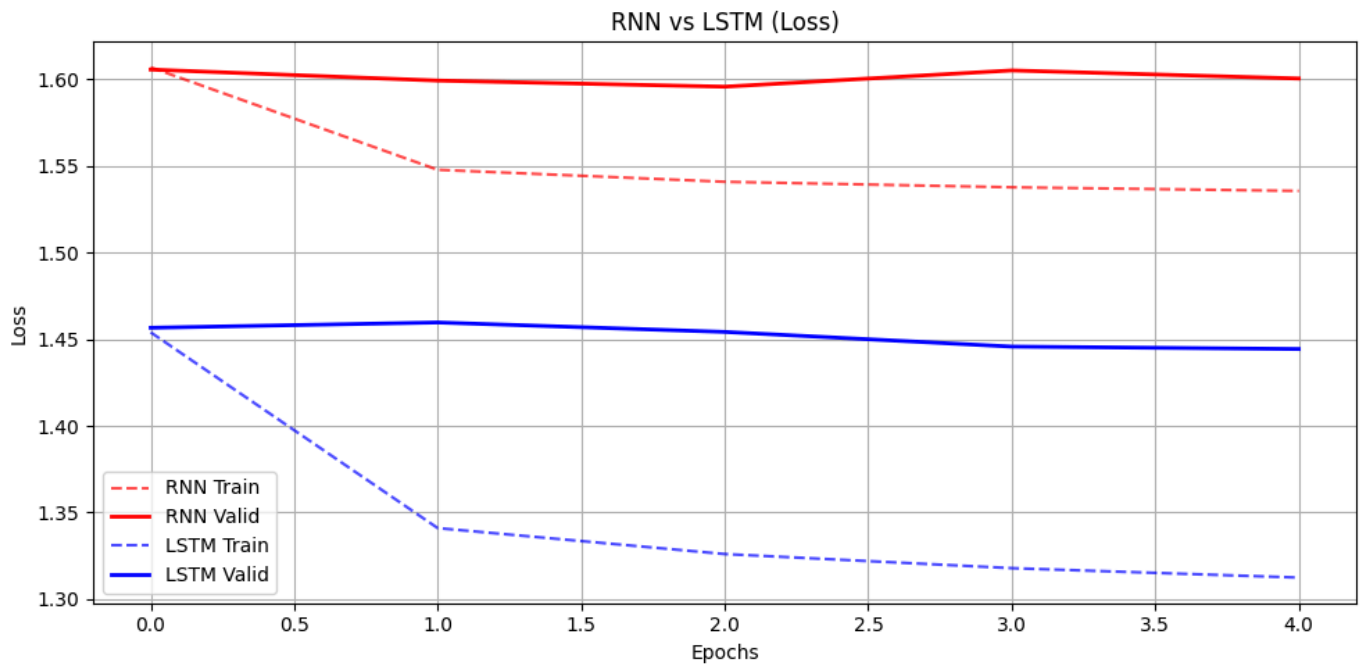
Observation (整體)：RNN 與 LSTM 都跑 5 epoch；LSTM 在相同設定下收斂更快、指標更好，尤其 hidden size 擴充到 256 時最明顯，以下逐段詳述。

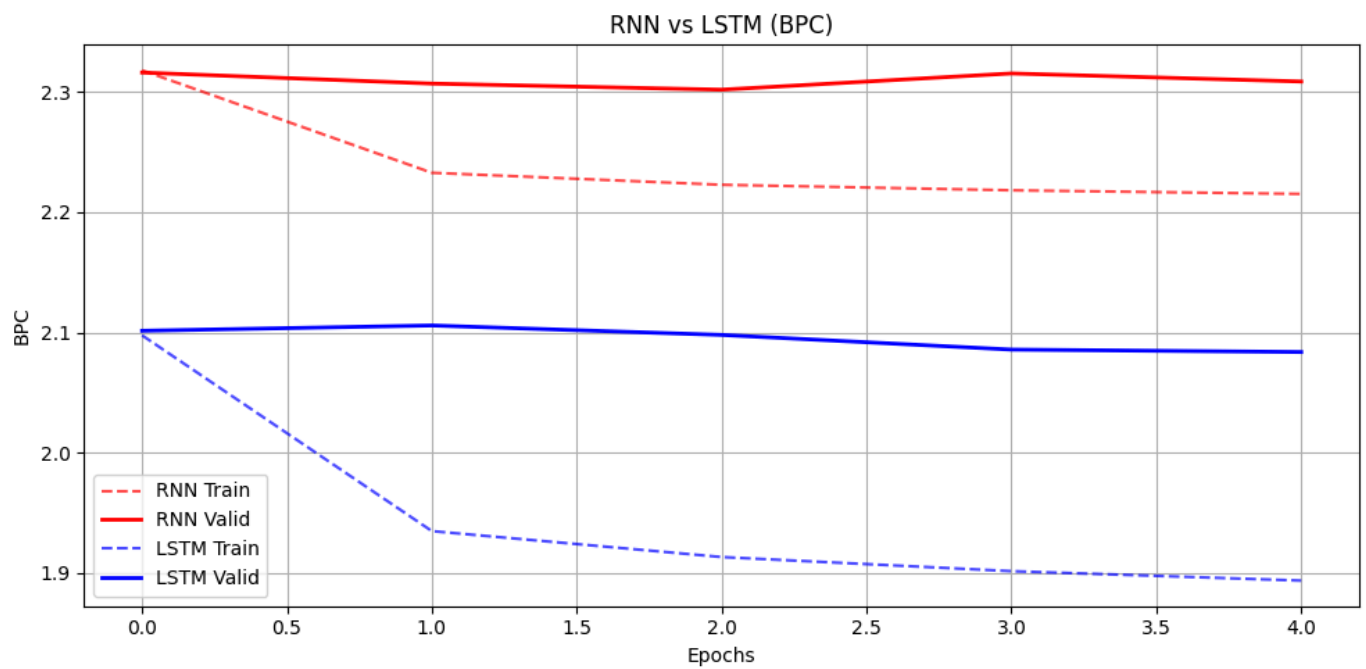
1. Standard RNN

1.1 架構與設定

Embedding → RNN(hidden=128, layers=2, dropout=0.25, batch_first) → Linear → vocab。

1.2 學習曲線與指標（RNN vs LSTM 對照）





RNN baseline (seq_len=100, hs=128) :

Epoch	Train Loss	Train BPC	Train Acc	Valid Loss	Valid BPC	Valid Acc
best (1)	1.6195	2.3364	0.5137	1.6089	2.3211	0.5210
last (5)	1.5532	2.2408	0.5297	1.6106	2.3236	0.5228

Observation (RNN 指標)：最佳在第 1 epoch，之後 valid loss 幾乎不變、acc 只有 0.52 左右，BPC ~2.32，顯示模型快到頂且能力有限。

1.3 5 個訓練截點生成 (RNN, 溫度=0.6, seed="The ", 長度=100)

[Epoch1] The sense the king with a cleak to a father were when the world.

GLoucester:

My leave of the grow her b

[Epoch2] The great than the set thee hence.

First Corent;

Let the time you to the hands, when have brave me, the

[Epoch3] The enterpain and so mistress and this are and scholer to say the wind of the brother's:

[Epoch4] The answer and grace and both of the rest.

DESDEMONA:

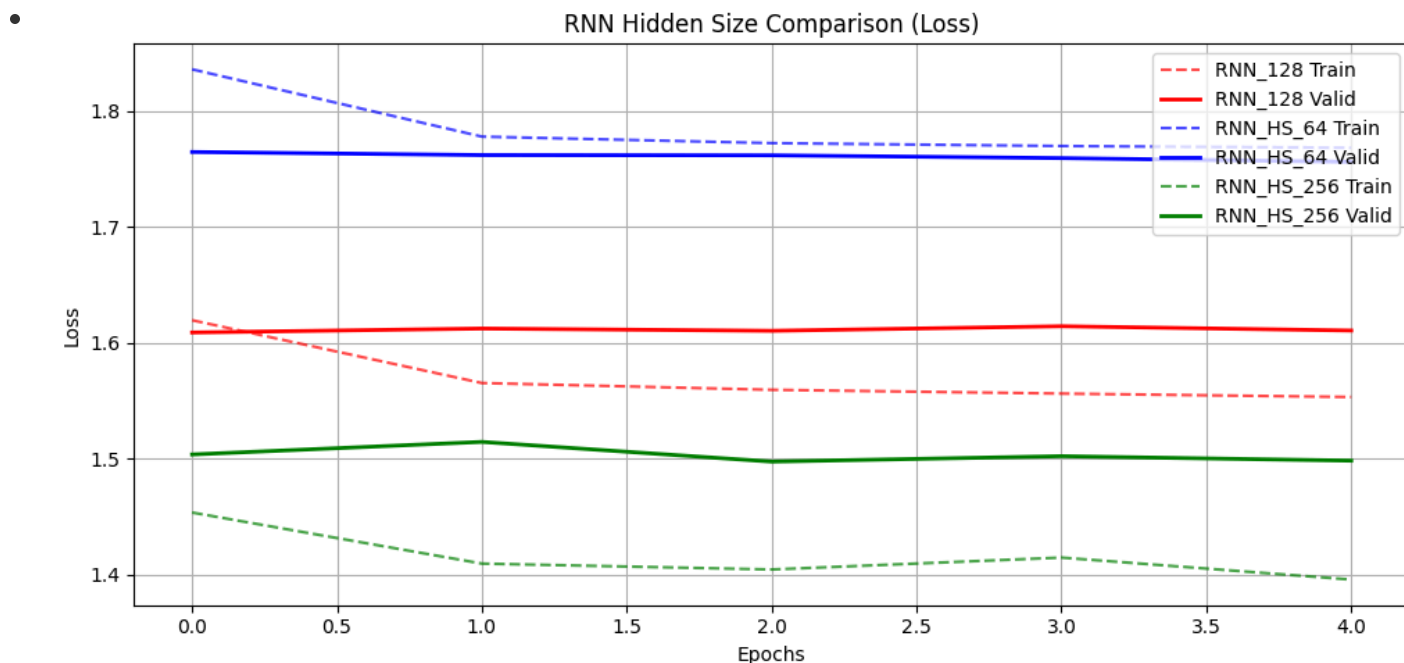
She were here that shall stand, that where the lo

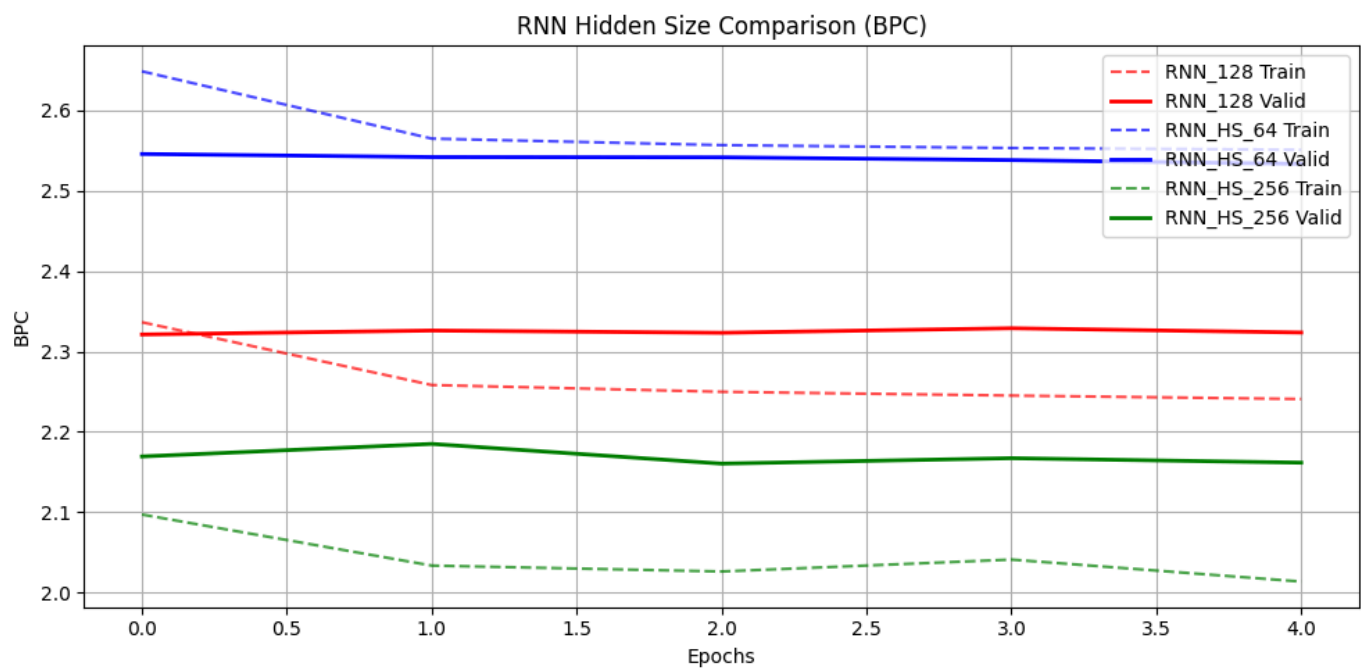
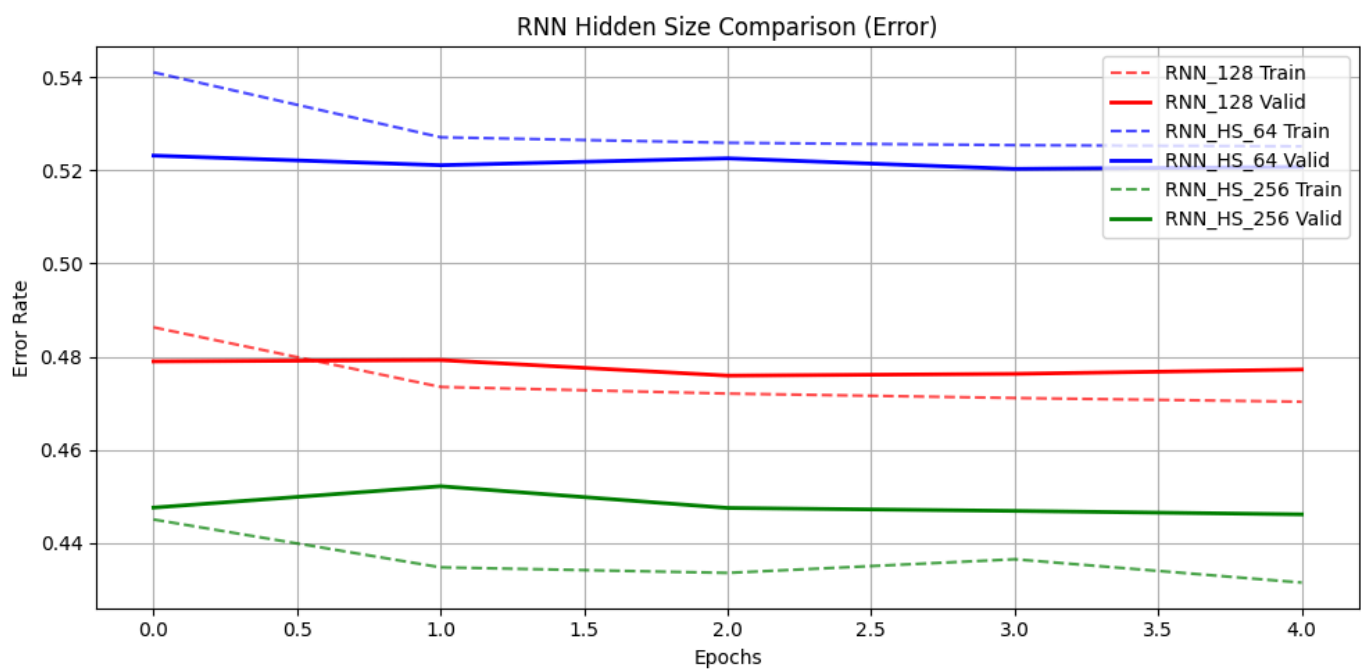
[Epoch5] The for the law to the consent, by this sons I am seem that they sent of the death, dei

Observation (RNN 生成)：早期假詞、斷句多；第 3–4 截點開始出現較長片段，但仍有缺字與怪詞；第 5 截點可讀性稍好但雜訊依舊。RNN 對長程依賴掌握有限。

1.4 RNN 超參數比較

Hidden size (seq=100)：

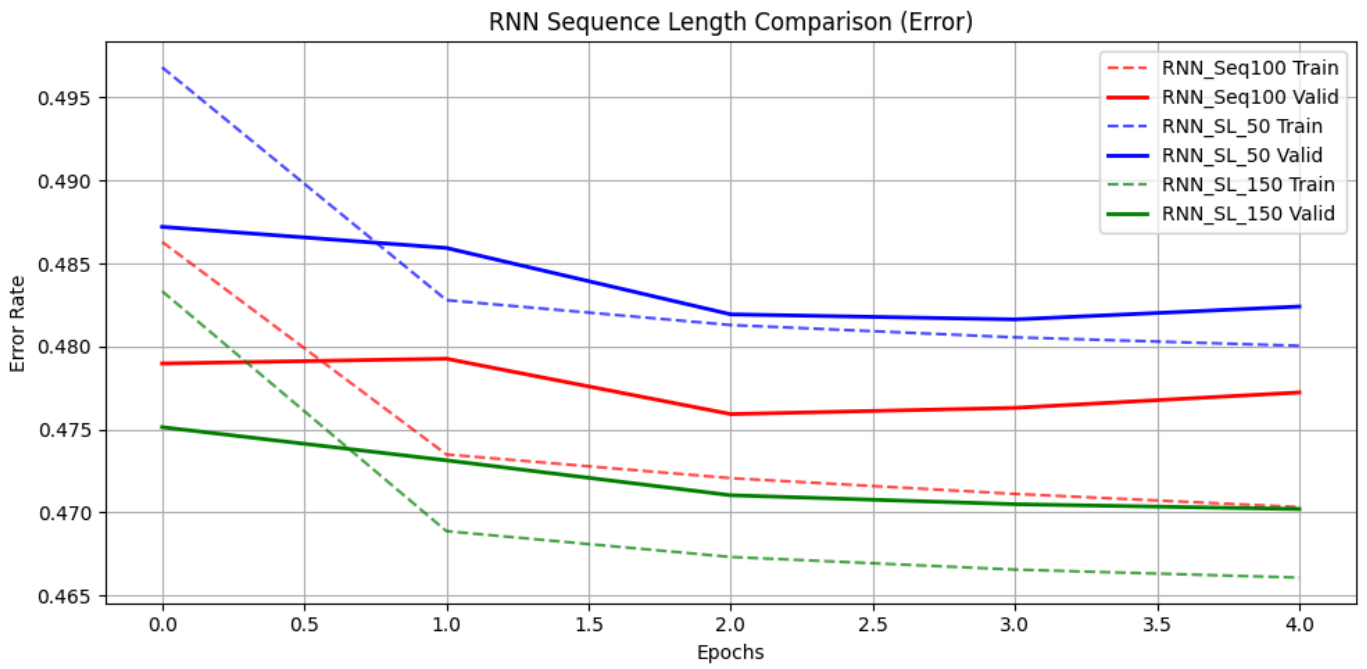
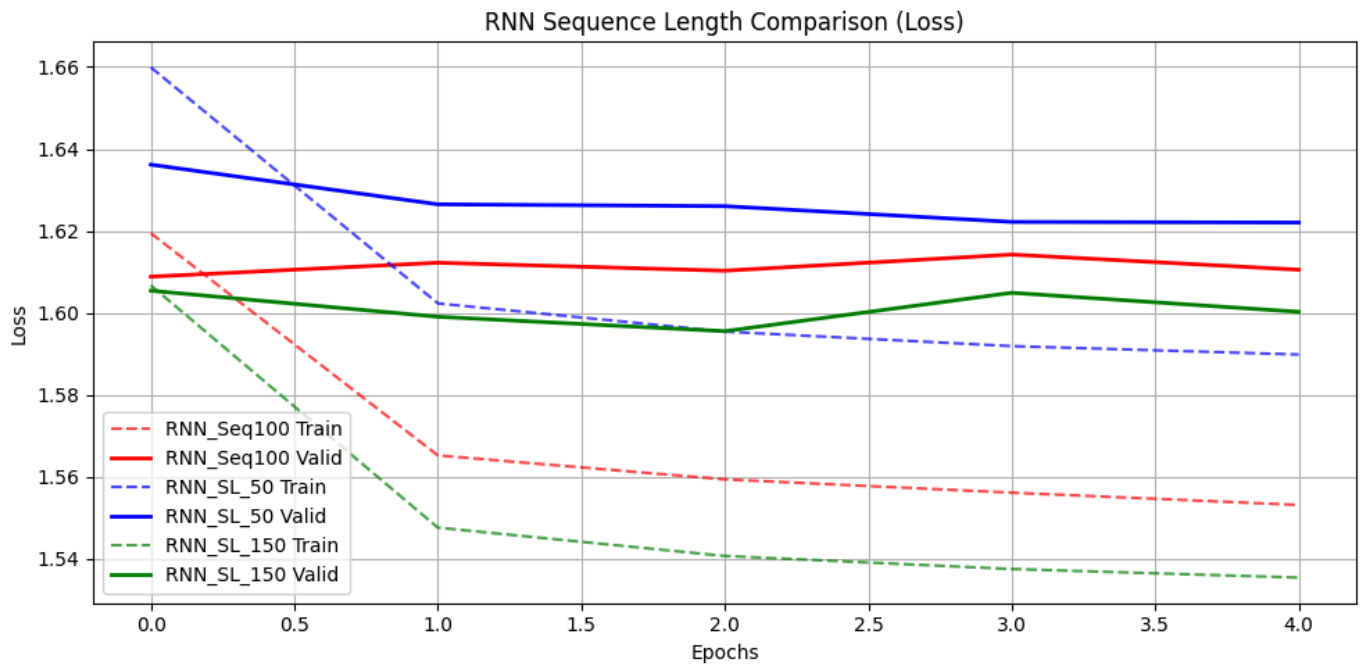


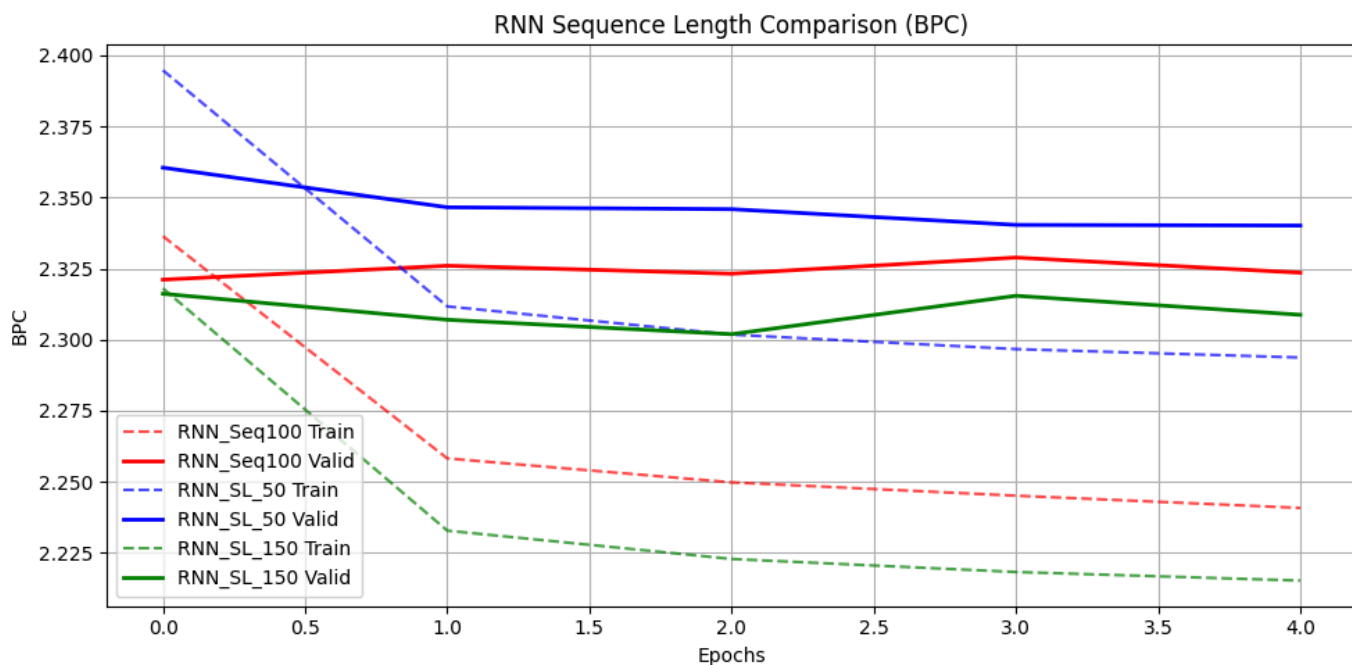


最佳驗證 loss：hs64=1.7561(e5), hs128=1.6089(e1), **hs256=1.4975(e3)**。

Observation：hs256 提升最明顯；hs64 欠擬合；隨 hidden 變大，loss/err/BPC 三條曲線都下移，但仍不及 LSTM。

Sequence length (hs=128)：





最佳驗證 loss：seq50=1.6221(e5), seq100=1.6089(e1), **seq150=1.5956(e3)**。

Observation：seq150 略優於 seq100，seq50 最差；拉長序列幫助有限，仍受 RNN 結構瓶頸。

2. LSTM（重做 1–3 並比較）

2.1 架構與設定

Embedding → LSTM(hidden=128, layers=2, dropout=0.25) → Linear。

2.2 學習曲線與指標

同上圖（RNN vs LSTM），LSTM 曲線顯著低於 RNN。

LSTM baseline (seq_len=100, hs=128)：

Epoch	Train Loss	Train BPC	Train Acc	Valid Loss	Valid BPC	Valid Acc
best (3)	1.3468	1.9430	0.5821	1.4423	2.0808	0.5698
last (5)	1.3344	1.9251	0.5851	1.4538	2.0974	0.5692

Observation（LSTM 指標）：驗證 loss 比 RNN 低約 0.17，acc 高約 4–5%，BPC 也顯著下降；在 3–5 epoch 即穩定，收斂較快。

2.3 5 個訓練截點生成 (LSTM, 溫度=0.6, seed="The ", 長度=100)

[Epoch1] The house of this fair plague, as thou shalt not father.

LUCY:

Let me think it is so spoke more to him.

[Epoch2] The lady so life and love him in her,

So banish'd in my brother with their seas,

Some more from the fres

[Epoch3] The company of his master's reserved,

Appointed the wind, and all the conscience

Is sworn before the Fre

[Epoch4] The counsel will be a patience that is awhile;

So play me and with my heart from her image.

COMINIUS:

C

[Epoch5] The Duke of Salisbury and Montague was,

I will be false to make to the bond.

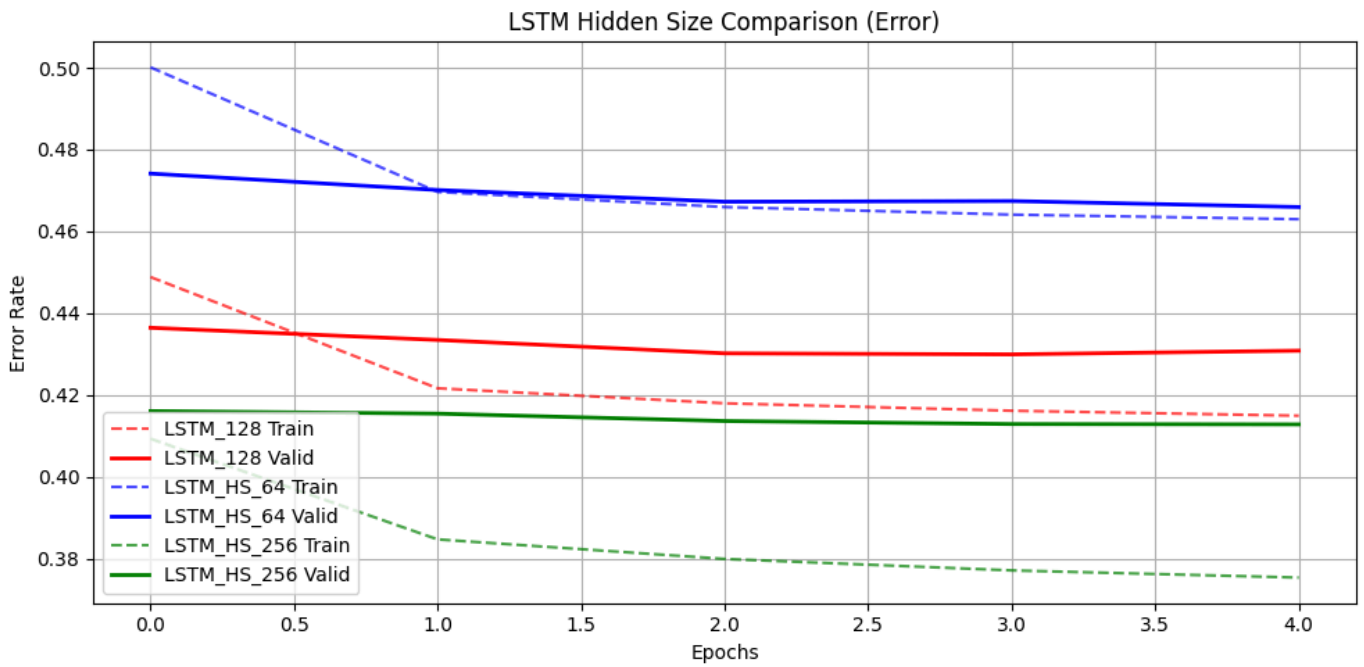
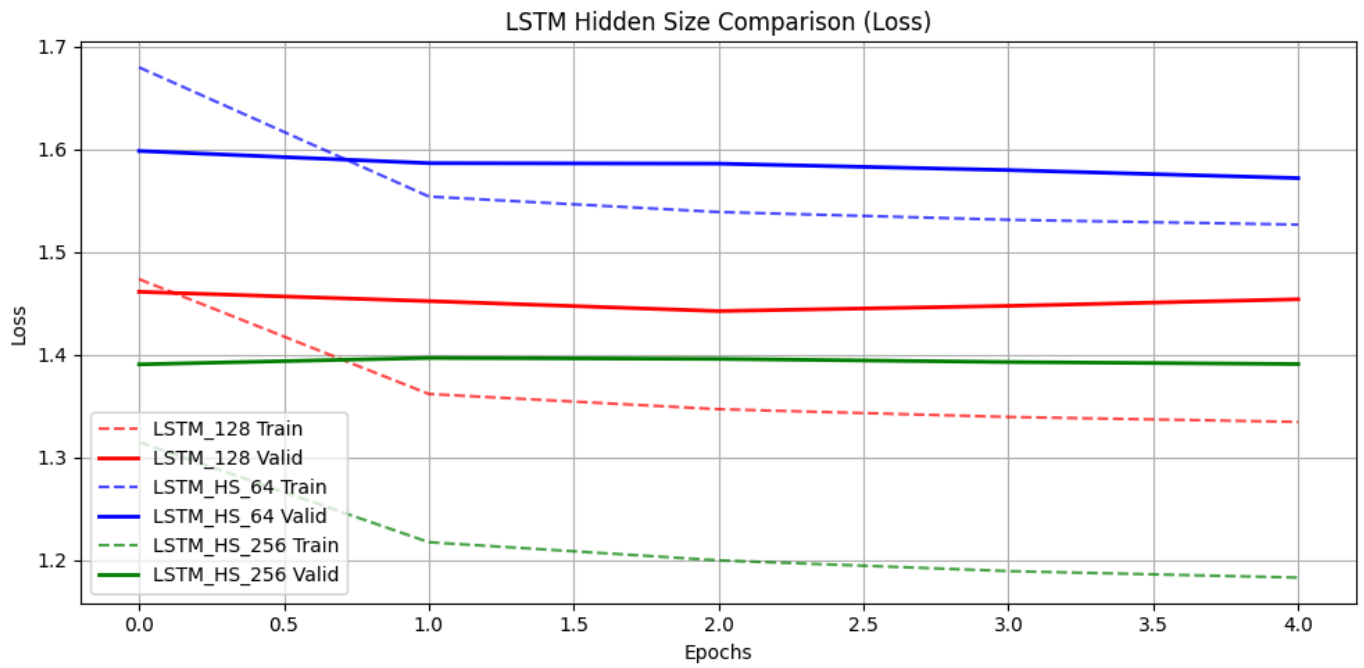
TOUCHSTONE:

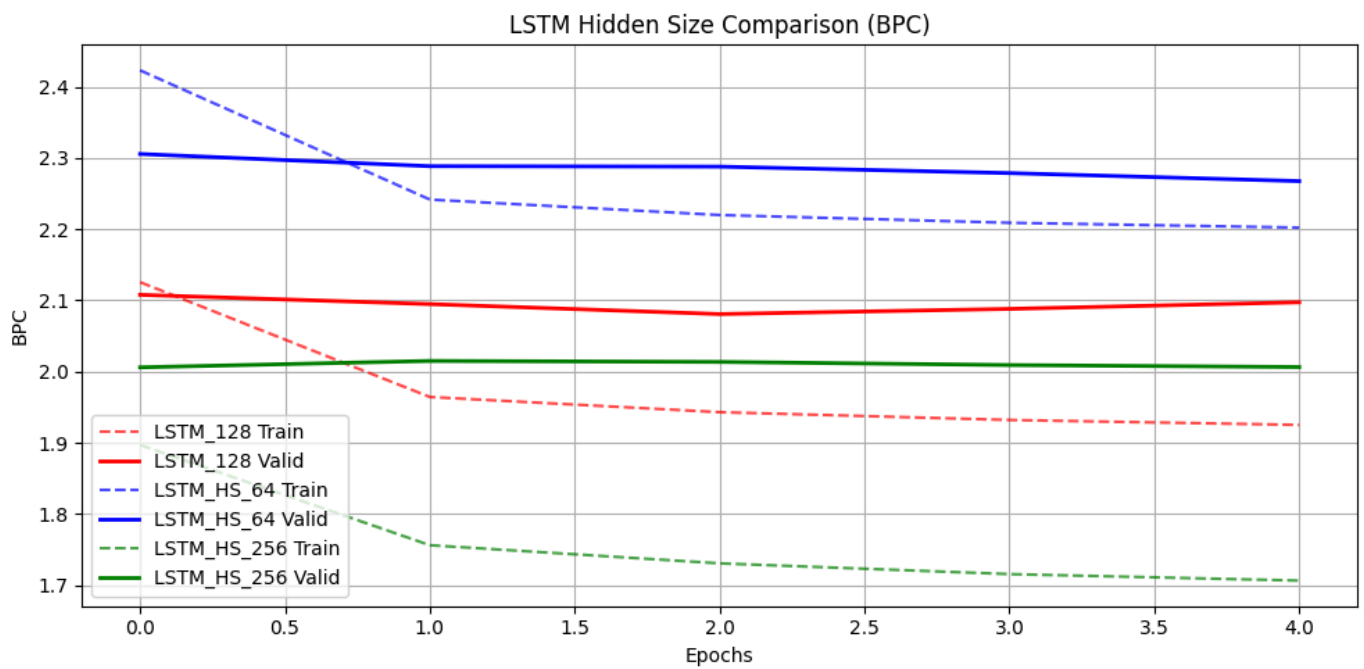
I think there

Observation (LSTM 生成)：第 1 個截點就能產生可讀台詞；第 3、5 截點句法與語意更自然，角色名常出現，連貫度明顯優於 RNN。

2.4 LSTM 超參數比較

Hidden size :

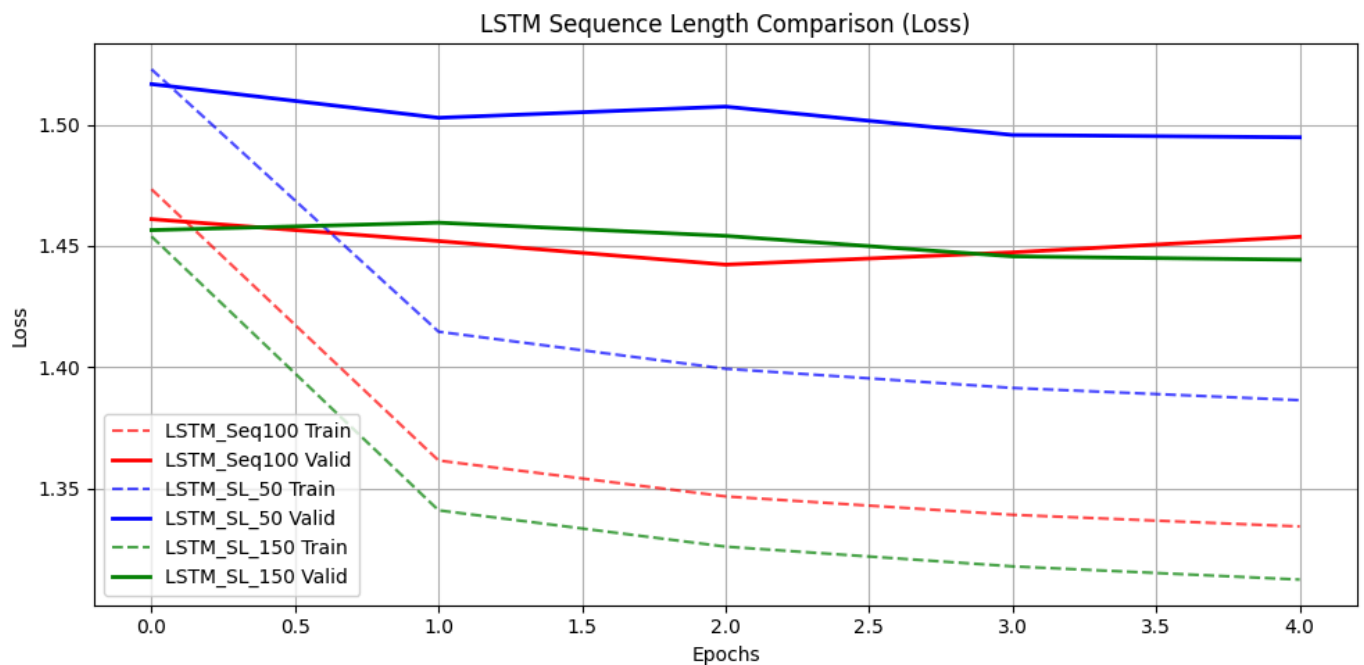


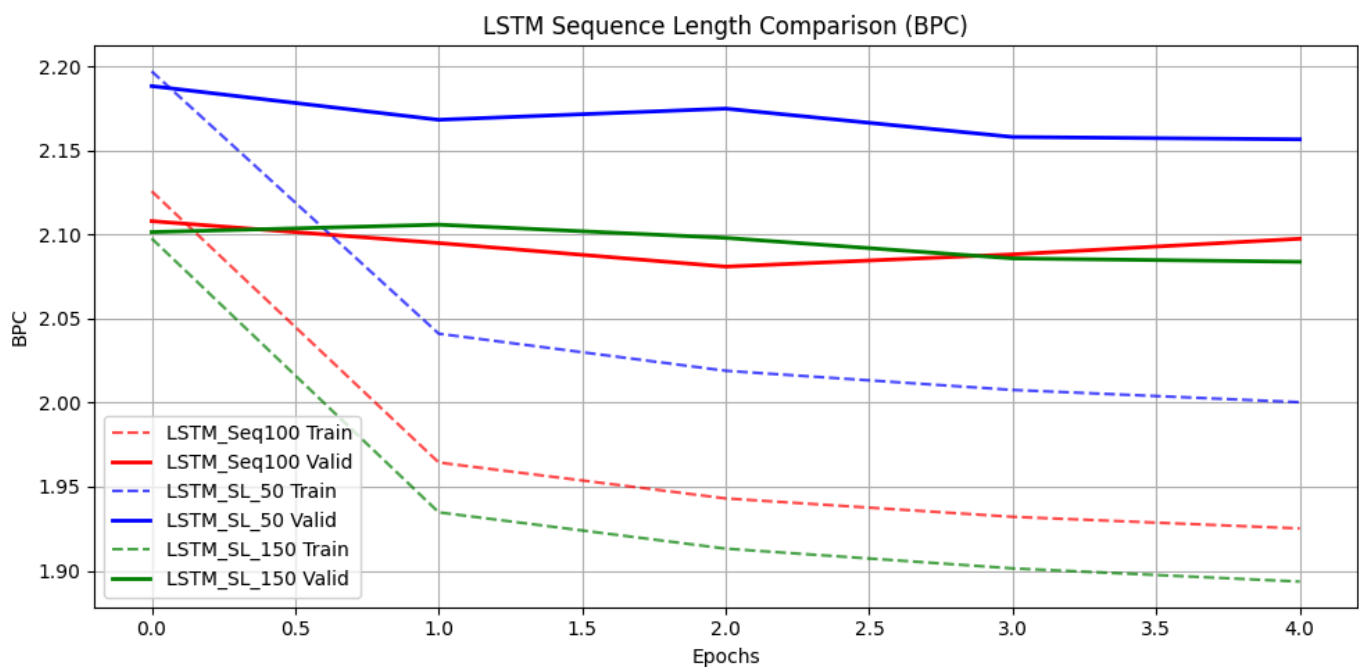
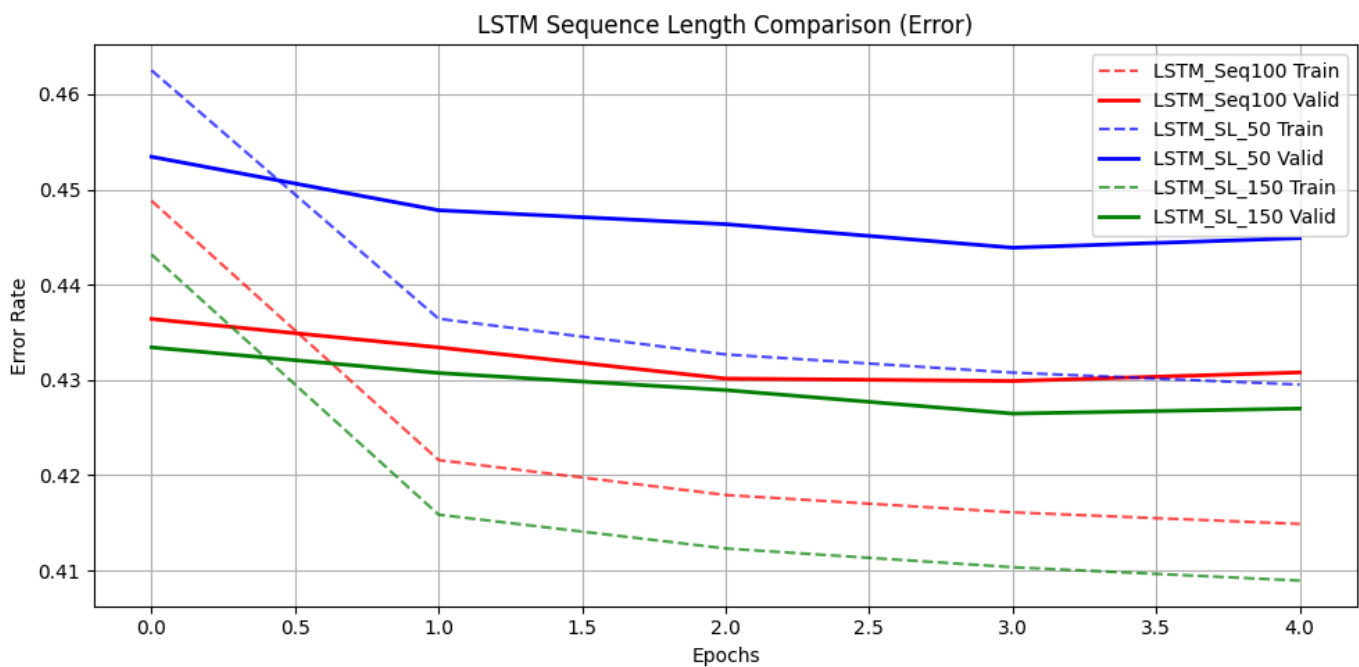


驗證 loss：hs64=1.5718(e5), hs128=1.4423(e3), **hs256=1.3905(e1)**。

Observation：hs256 在第 1 個 epoch 就達最佳，容量足且收斂快，因此我會鎖定這組。

Sequence length：





驗證 loss：seq50=1.4948(e5), **seq100=1.4423(e3)**, seq150=1.4443(e5)。

Observation：seq100 與 seq150 差異極小，seq50 稍差；考慮效率，我會選擇固定 seq100。

2.5 RNN vs LSTM 總結

- LSTM 數值全面勝出，尤其 hs256：驗證 loss=1.3905、BPC≈2.006、acc≈0.587（5 epoch）。
- Breakpoints 顯示 LSTM 句子流暢、劇本風格明顯；RNN 假詞與斷句較多。
- 收斂速度：LSTM 3–5 epoch 即趨平；RNN 拉長 epoch 也難有突破。

3. 文字生成 (Prime, 以最佳模型 LSTM_HS_256)

模型：LSTM_HS_256 (seq_len=100) , checkpoint checkpoints/best_model_LSTM.pth 。

檔案： outputs/generation_final.txt , Prime="JULIET" , 溫度=0.6 , 長度 500 (格式化為 12 行) 。完整輸出：

JULIET:

What, sir?

MARK ANTONY:

A heart, I

say, you will not a man forth to Cieson

to

prove my beauty, and his majesty of a worl

d.

TITUS ANDRONICUS:

Come, go with me.

CA

SSIO:

Here, my lord, and the bride-balls an

d moonlight

Of despite of Edward will be de

fend

The tribune to the Fulster's father to

the

fault of way.

LEONATO:

I cannot be no

t before the former than they know

The stoc

ks of heaven, and think he be made you and

the glory be the end: and the sea was provo

ked before

The danger of a party

Observation：全篇保持劇本格式，角色名頻繁且對話段落清楚；雖仍有少數斷行與不通順詞，但溫度 0.6 下整體穩定度佳、語氣一致性不錯，較高溫時的雜訊少。

4. 結論

- **最佳模型**：LSTM_HS_256 (seq_len=100)，驗證 loss=1.3905、BPC $\approx$ 2.006、acc $\approx$ 0.587 (5 epoch)。成本效益最佳。
- **是否要再訓練更久？** 這組在第 1 個 epoch 就接近最優，後續收益有限；其他組合差距大，即使拉長也難追上，我維持現狀。
- **生成溫度**：本次 prime 用 0.6，穩定度高；若要更多樣性可微調，但目前設定已符合需求。

所有圖片在 *plots/*，完整指標與生成輸出在 *outputs/*，*checkpoint* 於 *checkpoints/*。如需額外細節可直接檢視檔案。